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Geomagnetic changes across the last reversal recorded in lava flows from La Palma, Canary Islands

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Abstract. Two adjacent sequences of 41 and 21 individual lava flows have been sampled in the island of La Palma. Although no detailed correlation can be established between the two sections, they both encompassed the Matuyama-Brunhes (M-B) reversal. Stepwise alternating field (AF) and thermal demagnetization in air and in vacuum accompanied by investigations of the rock magnetic properties allowed us to identify several lava flows with the same signature and thus most likely associated with very short episodes of intense volcanic activity. The intermediate virtual geomagnetic pole (VGP) positions do not cluster within any preferred geographic areas. The polarity change is followed by a rebound (or an excursion) with a cluster of VGPs over northeastern India. Whether or not this feature is truly associated with the transition or correlated with the Delta event (0.69 Ma) will hopefully be resolved by further dating. Paleointensity experiments were performed on 60 samples using the modified Thellier method [*Coe*, 1967]. Corrections for magnetomineralogical changes induced by heating were applied using the technique of partial thermoremanent magnetization (pTRM) checks [*Valet et al.*, 1994b] with a significant increase in the success rate (from 38% to 58%). Twenty-four independent estimates of the paleofield range between 3.9 and 46.0 μT. The field intensity was reduced to at least one tenth of the present-day value during the transition. Low (transitional) field intensities are also observed during the rebound (or excursion). The lava flows preceding the polarity change show evidence for a significant drop in the dipole field intensity, while the posttransitional flows exhibit high field intensities. These features are compatible with the asymmetrical saw-tooth pattern recently observed in sedimentary records [*Meynadier et al.*, 1994]. The characteristics of the M-B transition emerging from a compilation of the four available volcanic records are not inconsistent with the hypothesis of rapidly time-varying nondipole transitional components favored by the decrease of the dipole and its subsequent recovery in the opposite polarity state.

Introduction

The last geomagnetic field reversal is the best documented in terms of the number of paleomagnetic records, most of which are from sedimentary records. A compilation of 21 records of the Matuyama-Brunhes (M-B) transition led *Clement* [1991] to propose that the virtual geomagnetic poles (VGP) were distributed in two preferred bands of longitude, one centered over South America and the other over Australia and eastern Asia. *Clement* [1991] also noted that this bimodal distribution coïncides with the longitudinal positions of primary concentrations of magnetic flux emanating from the core. *Gubbins and Kelly* [1993] recently claimed that the source of these flux features may have persisted over the past 2.5 Myr. From sedimentary records of Cenozoic reversals *Tric et al.* [1991] deduced the same geographical preferences of the VGP trajectories, and *Laj et al.* [1991, 1992] noted further a remarkable correlation between the preferred longitudes and north-south bands of anomalously fast *P* wave velocities in the lower mantle. Subsequent compilations and additional recent paleomagnetic records showed that the earlier analyses of VGP paths of the M-B transition were biased by the absence of normalization [*Valet et al.*, 1992], that some of the most detailed and famous records were in fact the results of artifacts in the magnetization process [*Larson and Patterson*, 1993; *Kent and Schneider*, 1994], and that the poor distribution of the observation sites is a major restriction to such compilations. Indeed, many VGPs paths were shown to be clearly dependent on the site longitude, an observation which is at least as significant as the existence of preferred transitional VGP paths [*Valet et al.*, 1992; *McFadden et al.*, 1993; *Quidelleur and Valet*, 1994]. It has also been suggested that the longitudinal confinement of transitional VGPs observed in most sedimentary records could reflect the presence of significant smoothing of the posttransitional and pretransitional field configurations [*van Hoof and Langereis*, 1991; *Langereis et al.*, 1992; *Kent and Schneider*, 1994].

In contrast to sediments, volcanic records do not display any longitudinal confinement of the VGPs [*Valet et al.*, 1992]. Analyzing records obtained from sequences of lava flows, *Hoffman* [1992] identified several clusters of VGPs about particular localities which he interpreted as evidence for long stands of the reversing field in inclined dipolar states. Subsequent compilations of all transitional volcanic VGPs led *Prévot and Camps* [1993] to reject the hypothesis of any preferred configuration of the transitional field. Only three records of the M-B reversal have been obtained from volcanic sequences [*Chauvin et al.*, 1990; *Baksi et al.*, 1992; *Brown et al.*, 1994]. Although rather poorly defined in terms of directional changes, these records show a clear discrepancy with those derived from

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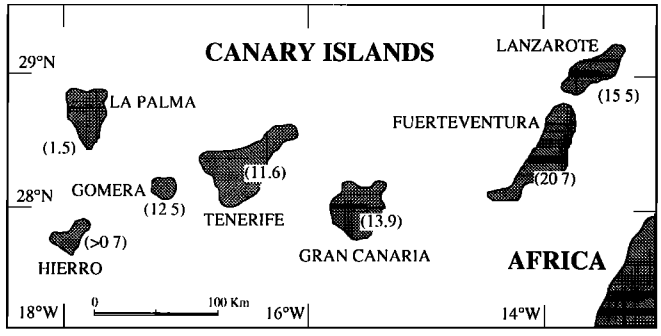


Figure 1 a. Map of the Canarian Archipelago including the oldest radiometric ages from subaerial volcanics in each island, reprinted from *J. Volvanol. Geotherm. Res.*, 60, Carracedo, J.C., The Canary Islands: An example of structural control on the growth of large oceanic-island volcanoes, 225-241, 1994 with kind permission of Elsevier Science - NL, Sara Burgerhartstraat 25, 1055 KV Amsterdam, The Netherlands.

sediments. However, except for two transitional flows on Tahiti [*Chauvin et al.*, 1990], none of these studies produced detailed paleointensity records. Analyses [*Mary and Courtillot*, 1993] based on the evolution of the full paleomagnetic vector across the reversal boundary demonstrate that further documentation of field intensity variations is crucial in constraining the mechanisms associated with reversals. Recent data sets from sedimentary cores show evidence for asymmetrical saw-tooth patterns associated with a succession of reversals [*Valet and Meynadier*, 1993; *Valet et al.*, 1994a; *Meynadier et al.*, 1994]. In these cores, the intensity decreases slowly during intervals of stable polarity and recovers rapidly immediately after the transition. These features, observed in three different oceans, should be present in volcanic records, provided that a detailed chronology can be established .

The aim of this study is to provide a new record of the field variations that occurred across the M-B reversal, with a special emphasis on intensity changes during the pretransitional and posttransitional periods. Following preliminary analyses by *Abdel-Monem et al.* [1972] and L. Tauxe (personal communication, 1993), our work focused on two adjacent sequences of lava flows on the island of La Palma (Canary Islands).

Sections and Samplings

The Canarian archipelago is composed of seven major volcanic islands (Figure 1). The western islands lie above oceanic crust, but the central part of the archipelago is underlain by a

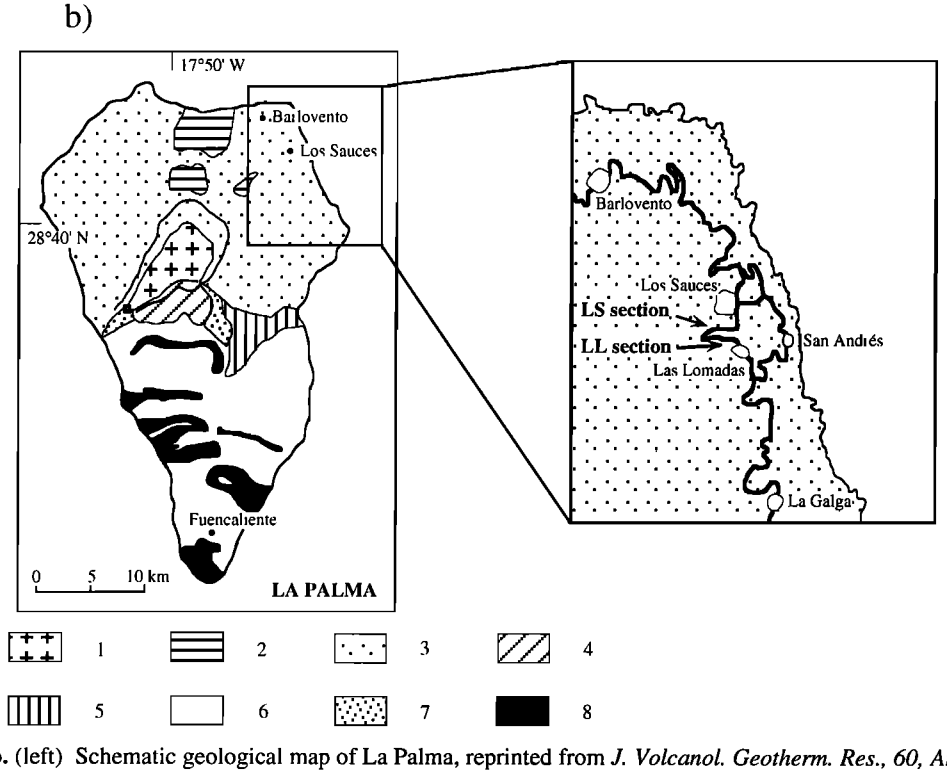


Figure 1 b. (left) Schematic geological map of La Palma, reprinted from *J. Volcanol. Geotherm. Res., 60, Ancochea et al.*, Constructive and destructive episodes in the building of a young oceanic island, La Palma, Canary Islands, and genesis of the Caldera de Taburiente, 243-262, 1994 with kind permission of Elsevier Science - NL, Sara Burgerhartstraat 25, 1055 KV Amsterdam, The Netherlands.: 1, Basal complex; 2, Lower Old Series (LOS); 3, Upper Old Series (UOS); 4, Bejenado massif; 5, Cumbre Nueva Series (CNS); 6, Cumbre Vieja Series (CVS); 7, sediments; 8, historical eruptions. (right) Location of the Los Sauces (LS) and Las Lomadas (LL) sections with reference to the main road.

transitional zone between oceanic and continental crust [*Dash and Bosshard*, 1969]. The oldest volcanic rock (38.6 Ma) and the oldest subaerial flow are found at Fuerteventura [*Carracedo*, 1994]. Volcanism has persisted more or less continuously during the last 20 Myr [*Abdel-Monem et al.*, 1971] and is still active on several islands; the most recent eruption occurred in 1971 at the Teneguia volcano in the southern part of the island of La Palma. The island of La Palma was built on Jurassic oceanic crust in the northwestern part of the archipelago [*Kent and Gradstein*, 1984]. The highest point of the island rises approximately 6500 m above seafloor (about 2400 m above sea level). The geology, which was described in detail by *Staudigel and Schmincke* [1984] and more recently by *Ancochea et al.* [1994], is characterized by two major structural units: a Pliocene submarine and plutonic complex in the lower part, and Pleistocene subaerial lavas which cover most of the island [*Afonso*, 1974]. The more recent unit (Pleistocene) in the northern edifice is composed of the Lower Old Series (LOS) and the Upper Old Series (UOS). By correlation with biostratigraphic levels an age of about 3-4 Ma was deduced for the oldest volcanic rocks [*Staudigel*, 1981]. Potassium-argon dating by *Abdel-Monem et al.* [1972] gave an age of 1.61 Ma for the oldest lava flow of the UOS .

The two sections involved in the present study were sampled in the UOS during two field trips in June 1993 and February 1994 (Figure 1b, right). A total of 494 cores were drilled with a portable gasoline-powered drill and oriented with a magnetic compass. Whenever possible, Sun orientation was preferred. Since the magnetic azimuths were consistently found to be 5° lower than the solar azimuths, we applied a correction of 5°W to the magnetically oriented samples. The dip of the lava flows, when measurable, ranged between 0 and 40° with azimuths between 300°E and 0°.

The section of Las Lomadas (which we will refer to as LL) is the south rim of the canyon "Barranco de Las Lomadas" (Figure 1b, right). K/Ar dates obtained by *Abdel-Monem et al.* [1972] were recalculated by *Ancochea et al.* [1994] with the decay constants of *Steiger and Jäger* [1977]. The results indicate ages of 0.83 ± 0.09 Ma and 0.78 ± 0.09 Ma for the lowest (LL141) and highest (LL101) flows, respectively. We note that these two values bracket the age of the M-B polarity reversal. The flows are less than 1-10 m thick and in many cases appear to be separated only by scoria, thereby suggesting high extrusion rates, which are characteristic of the Canary Islands [*Carracedo and Soler*, 1995]. However, we did find several baked paleosols (Figure 2). Occasionally, breccias were observed, and there are thick vegetal covers between flows LL139-LL140 and LL128-LL129.

The second section (Los Sauces, LS) is located on the other side of the canyon (Figure 1b, right) and includes 21 successive lava flows (Figure 2). No radiometric age was obtained for this section, but 10 lava flows had been previously sampled by L. Tauxe and coworkers. Despite the absence of detailed paleomagnetic studies, preliminary analysis of these samples had revealed the existence of a polarity reversal that could reasonably be identified as the M-B transition. With the help of these results and information that was kindly provided to us by L. Tauxe, we could recognize the successive units and their polarity. Unfortunately, this sequence is interrupted by about 100 m of vegetal cover between LS107 and LS118. We thus sampled a vertical succession of 10 lava flows outcropping immediately above LS118 and numbered from LS108 (at the top) to LS117. Despite the lack of direct correlation, field observations suggest that LS108 is stratigraphically lower than LS107.

The lowest flow from the LS section (LS121) is close and approximately at the same elevation as the lowest flow from the LL section (LL141), which implies that both sections have very similar ages. However, the correlation of the two sections is not straightforward (Figure 2). The thin baked soils can result from very local phenomena and therefore cannot be used for correlations over hundreds of meters. In fact, their presence cannot even be taken as an indication of a large time gap between two successive flows [*Chauvin et al.*, 1990]. In contrast, the pyroclastic deposit near the top of the LS section (Figure 2) should be present over a wide area. The fact that this level was not observed in the LL section suggests that the LL section is older

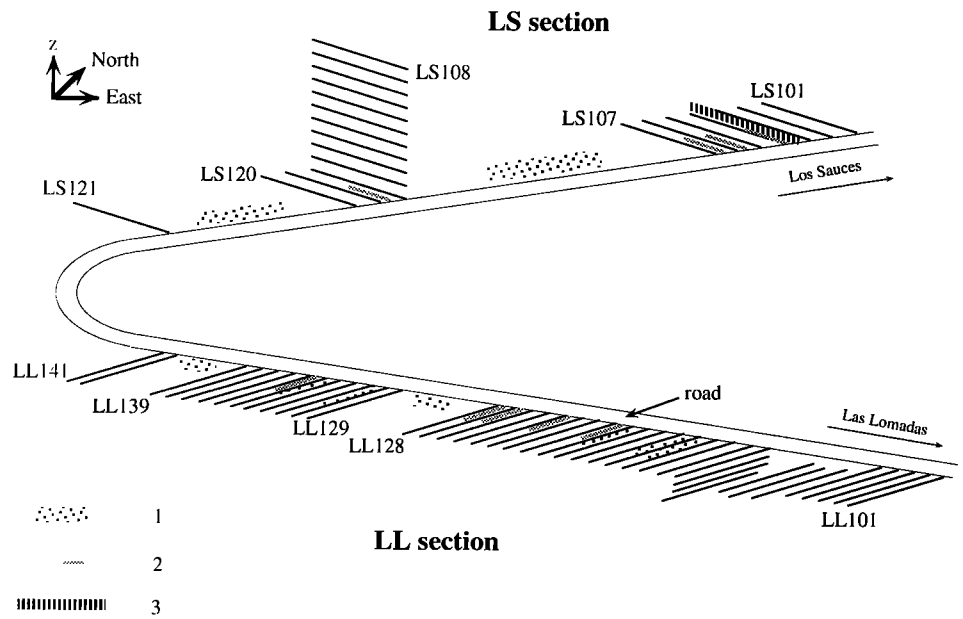


Figure 2. Schematic description of the stratigraphy of the LS and the LL volcanic sequences showing the succession of the sampled flows. The locations of 1, vegetal cover and/or volcanic breccia; 2, baked paleosoil; and 3, pyroclastic flow are also shown.

than the upper three flows of the LS section. Interestingly, long but narrow lava flows are typical of the recent volcanism in the southern part of the island. There is thus no reason why two sequences separated by a few hundred meters should be identical. Such considerations, in addition to the absence of unambiguous stratigraphic indicators, convinced us not to attempt any detailed correlation.

Laboratory Experiments and Techniques

Natural Remanent Magnetization (NRM)

Measurements and demagnetization of the NRM were performed in the shielded room of the Institut de Physique du Globe de Paris (IPGP) paleomagnetic laboratory. At least two samples from each flow underwent a thermal treatment; in most cases the first one was in air and the other one was in vacuum. The measurements were processed with a Molspin or a JR5 (Geofizica Brno) spinner magnetometer. Two (or more) additional samples were used for alternating field (AF) demagnetization, performed with a 2G cryogenic magnetometer [*Nagy and Valet*, 1993]. Thus, for every flow we could compare the respective efficiency of three different cleaning methods. In most cases, AF and thermal demagnetization gave consistent results, but for some weakly magnetized samples, thermal demagnetization was more efficient. Furthermore, for flow LS112, thermal demagnetization performed under vacuum was the only procedure able to isolate the characteristic components. The low-field magnetic susceptibility was measured with a Bartington susceptibility meter after every thermal demagnetization step.

Magnetic Mineralogy

One or two samples from each flow were selected for investigation of magnetic properties. Anhysteretic remanent magnetization (ARM) was imparted in a 0.05 mT direct field and subsequently demagnetized. The *S* ratio typically used to estimate the relative amounts of magnetite and hematite was calculated from the isothermal remanent magnetization (IRM) acquired in a direct field of 1 T and from that acquired in a subsequent reversed field of 0.3 T ($S = -\mathrm{IRM}_{(-0.3T)}/\mathrm{IRM}_{(1.T)}$ [*Stober and Thompson*, 1979]). Following this experiment, a new $\mathrm{IRM}_{(1.T)}$ was stepwise AF demagnetized in peak fields of 10 mT and 30 mT. This procedure was repeated before thermal demagnetization at 350°C and 450°C. The low-field susceptibility (χ) was measured before and after every thermal demagnetization step. These experiments provide information about the magnetic mineralogy and granulometry of the samples.

Concentrations of magnetic minerals can be deduced from values of χ, ARM, and IRM. The upper part of the LL section shows high variability in concentration as reflected by large variations between successive lava flows in IRM and ARM (Figures 3a). The χ values oscillate over the entire section. Concentrations appear to be more uniform in the LS section (Figure 3b) as inferred from constant values of ARM and IRM; flow LS119 shows the highest magnetic content.

A first indication about the carriers of the natural remanent magnetization (NRM) can be deduced from the behavior of the samples upon stepwise thermal demagnetization. We identified four categories of behavior (Figure 4). The NRM of the type A samples is characterized by a strong resistance to demagnetization up to temperatures higher than 500°C, followed by a very rapid decrease with complete removal at 580°C. This behavior is consistent with magnetite in a narrow range of grain sizes. The type B samples show a more or less regular decay of the NRM up to 580°C, which indicates the presence of pure magnetite with different grain sizes. The type C samples exhibit two components of magnetization with Curie temperatures close to 400°C and 500-560°C. This behavior is probably due to the presence of titanomagnetite with varying Ti content. The last type of behavior is characterized by a soft component, probably titanomagnetite with a high Ti content, that carries 90% of the NRM and has a low blocking temperature (300-350°C). The corresponding high-temperature component decreases regularly until the Curie temperature of magnetite. In most cases these four categories are characteristic of a given flow and alternate within each section. In a few cases, two different types of behavior have been observed within a single flow; only a few samples did not fit unambiguously within a single type of demagnetization diagram. Since no Curie temperature higher than 580°C was detected, there is no indication of hematite. The values of the *S* ratio are mostly close to unity with a minimum of 0.94. This confirms that there are no high-coercivity minerals carrying the NRM. The evolution of the low-field susceptibility after 350°C and 450°C in the two sections gives no indication that maghemite was converted into hematite. Thus we can reasonably conclude that magnetite and titanomagnetite with varying titanium contents are largely dominant in both sections. Because the magnetization is due to magnetite, changes in magnetic grain sizes between successive lava flows can be depicted by ratios such as ARM/χ (Figures 3c and 3d) [*Banerjee et al.*, 1981], $\mathrm{IRM}_{10\,mT}$/IRM, and IRM_{30mT}/IRM (IRM_{10mT} and IRM_{30mT} represent the $\mathrm{IRM}_{(1.T)}$ demagnetized in peak AF of 10 mT and 30 mT, respectively). The type A demagnetization diagrams are associated with high values of IRM_{30mT} and the ARM/χ ratio, which confirms the presence of single-domain grains of magnetite with high coercivities. The type B samples show lower values of the IRM_{30mT}/IRM ratio and lower values of ARM/χ, indicative of lower coercivities. This is consistent with a wide range of grain sizes of magnetite as previously deduced from the demagnetization diagrams.

Determination of Absolute Paleointensity

The modified Thellier double heating method [*Coe*, 1967] was used for paleointensity experiments. Heating and cooling of batches of 20 samples were performed within a Pyrox oven [*Quidelleur et al.*, 1992] in the shielded room of the IPGP. Heating was performed under vacuum (less than 10^{-2} torr) and cooling was regulated by a stream of nitrogen gas precooled by circulation within liquid nitrogen. Temperature control was provided by three thermocouples within the inner tube of the oven, but accurate monitoring of the temperature of the samples was achieved with three other thermocouples positioned within the sample holder. Heating at 300°C and subsequent cooling to room temperature took about 3 hours, while 4.5 hours were necessary at 500°C. The same cooling rates were used for every experiment. The magnetic field was measured for every sample position along the holder and ranged from 18.9 μT to 44.1 μT for the different experiments.

The absolute field values were first calculated by using the conventional thermoremanent magnetization (TRM)-NRM diagrams where the TRM acquired at each temperature step is plotted against the NRM remaining at this temperature [*Nagata et al.*, 1963]. Ideal behavior would be that all the TRM-NRM points are aligned along a straight line. However, mineralogical changes are often induced during heating (or cooling) and are not

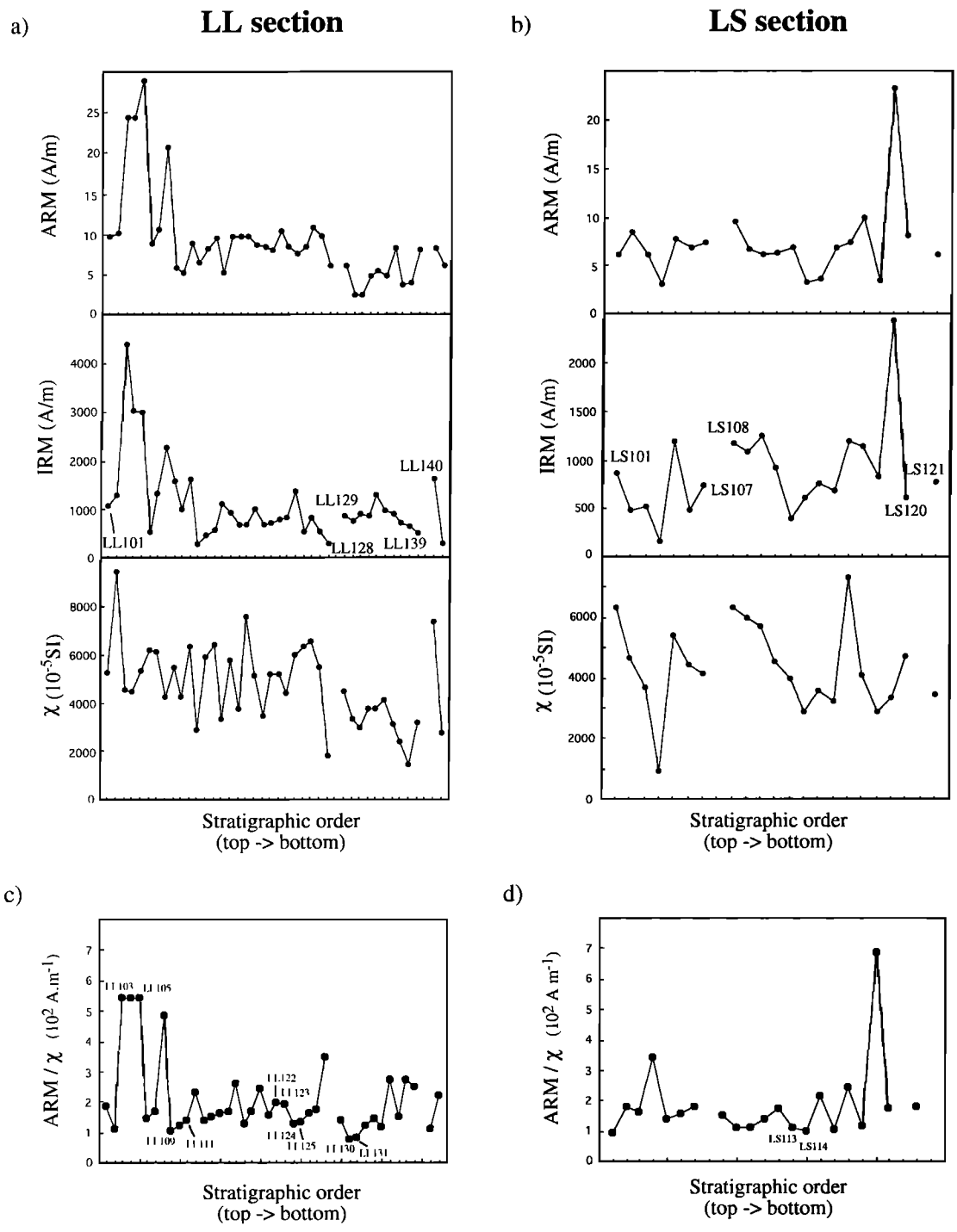


Figure 3. (a) Evolution of the ARM, IRM and magnetic susceptibility (χ) measured on one sample per flow from the LL section. The discontinuities represent the stratigraphic location of sampling gaps. (b) same as Figure 3a, but for the LS section. (c) Evolution of the ARM/χ ratio in the LL section. (d) same as Figure 3c, but for the LS section.

necessarily detectable as significant deviations from linear behavior [*Prévot et al.*, 1983]. The method of partial TRM (pTRM) checks has been shown to be a very effective (if not the most effective) technique for detecting such changes. These checks have been used in the past to select TRM-NRM diagrams suitable for paleointensity determinations. In principle, successive mineralogical changes induce successive loss or gain in the ability of a sample to acquire TRM, as illustrated by "negative" pTRM checks. *Valet et al.* [1994b] recently proposed a method to correct the TRMs for the successive deviations between the pTRM checks and the original TRM. The correction uses the residual (Res_{T_i}) between the pTRM check at $T=T_i$ (performed from T_{i+1} to T_i) and the original TRM value (at $T=T_i$) to obtain a corrected TRM for temperature T_{i+1}. In addition, to account for changes between T_i and T_{i+1}, we can extrapolate Res_{T_i} by relying on the assumption that the increase in Res_{T_i} between T_i and T_{i+1} is proportional to the increase in the altered TRM given by $TRM_{T_{i+1}}$ / $pTRMcheck_{T_i}$. The same additive law remains valid as in the

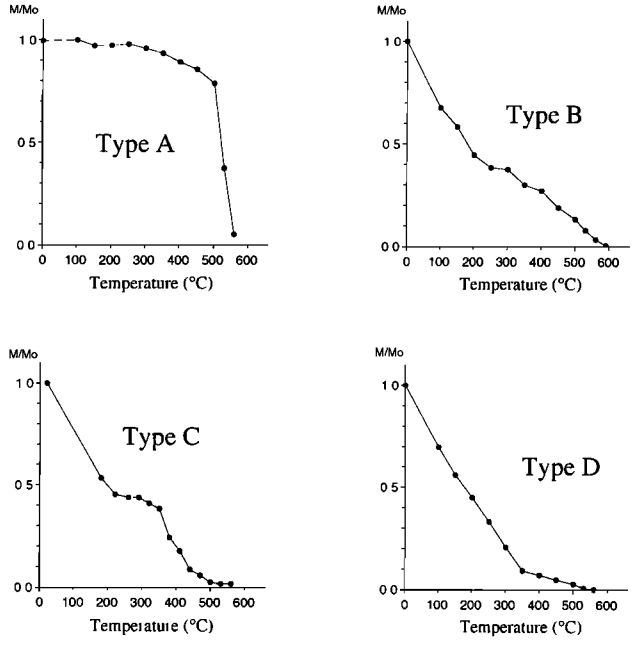


Figure 4. Typical thermal demagnetization diagrams of the natural remanent magnetization (M) normalized to the maximum value (M_0).

previous case. The values of TRM are thus calculated as follows:

$$TRM_{T_{i+1}} = TRM_{T_{i+1}} - Res_{T_i} \times (TRM_{T_{i+1}}/pTRMcheck_{T_i}),$$

with

$$Res_{T_i} = pTRMcheck_{T_i} - TRM_{T_i}.$$

The method requires at least three pTRM checks in order to obtain more than three corrected pTRM-NRM points, but obviously as many checks as possible are needed to reduce the uncertainties. Since the correction method was developed after the present experiments were completed, pTRM checks are not available at every temperature step. Because the correction implicitly assumes that most changes involve grains with low blocking temperatures, we have only retained samples that did not show significant differences between the corrected paleointensity values before and after extrapolation. Four samples which showed large inconsistencies between the NRM-TRM diagrams obtained before and after extrapolation were eliminated from the final result. For most samples, five pTRM checks were performed between 220°C and 500°C. It is important to note that because pTRM checks were not performed at every step, there is a smaller number of points in the corrected TRM-NRM diagrams than in the noncorrected diagrams. However, in most cases the temperature range used to determine the paleointensity value is wider. The pTRM corrections were calculated for every sample, with a significant increase in the rate of success: 23 out of 60 paleointensities (38%) could be calculated without any correction whereas 35 values (58%) were obtained after correction.

Results

Directions

Typical demagnetization diagrams from both sections for samples with normal, intermediate and reverse polarities are shown in Figure 5. Except for an overprint of the present field which was easily removed between 300°C and 400°C, many diagrams show univectorial behavior. Samples with intermediate directions are characterized by a lower magnetization intensity and consequently a larger viscous component. Both AF and thermal demagnetization were effective in isolating the Characteristic remanent magnetization (ChRM) for all the samples except flows LS112 and LL116 for which thermal demagnetization (performed under vacuum for LS112) was most useful. Difficulties in isolating the ChRM in these two flows are probably related to the low NRM intensities and the presence of secondary components. The ChRM were determined using least squares principal component analysis [*Kirschvink*, 1980] involving temperatures greater than 400°C.

The mean flow directions were calculated using *Fisher* [1953] statistics after rejecting a few outliers. The mean directions and their associated VGP are listed in Tables 1 and 2 for the LS and LL sections, respectively. Figures 6a and 6b summarize the declinations, inclinations, and VGP latitudes of the successive lava flows in both sections.

The lowest flow (LL141) of the LL section is 0.83 ± 0.09 Ma old [*Abdel-Monem et al.*, 1972] and has a reverse polarity consistent with the late Matuyama chronozone. Since the upper flow has a 0.78 ± 0.09 Ma K-Ar age [*Abdel-Monem et al.*, 1972], there is little doubt that the R-N transition recorded between flows LL129 and LL128 is the Matuyama-Brunhes reversal. Indeed, this time interval agrees with the new age determination of 0.78 Ma for this transition [*Shackleton et al.*, 1990; *Bassinot et al.*, 1994] recently obtained from astronomical calibration of isotopic records. Recent $^{40}Ar/^{39}Ar$ radiometric dating of Hawaiian transitional flows yields a similar result of 0.783 ± 0.011 Ma [*Baksi et al.*, 1992]. The section LL is characterized by an abrupt polarity change followed by a large excursion in declination and inclination (between flows LL118 and LL101). Unfortunately, the transition coincides with several tens of meters of vegetation cover. The onset of the transition shows a departure of about 40° from the axial dipole direction. The polarity change is also recorded in the LS section, but the results, when scrutinized in detail, show different characteristics. A single intermediate direction is present before the transition (LS118). In contrast to the LL section, the prereversal inclinations are steeper than the axial dipole field value at the site latitude. The reversal itself occurs between two superimposed flows (LL112 and LL111). There are no excursional directions following the polarity change.

The differences between the two sections probably reflect the discontinuous character of the volcanic eruptions. Another indication of such irregularities is suggested by the presence of nearly identical directions between several successive lava flows. It is thus reasonable to envisage that the eruptions were associated with active periods of the volcano. Usually, successive extrusions over a very short time interval show similarities in volcanic products. In the absence of detailed geochemical and mineralogical analyses, we attempted to identify successive flows on the basis of similarities in magnetic properties. Among the possible indicators, the ARM/χ ratio (Figures 3c and 3d) appears to be the most appropriate because highly reproducible values were obtained within individual lava flows. When considered with the different categories of demagnetization diagrams, this parameter allowed us to identify several groups of consecutive flows with the same properties (LL103/LL105; LL109/LL111; LL122/LL123; LL124/LL125; and last, LL130/LL131 for the LL section and LS113/LS114 for the LS section). Each group can reasonably be considered as single units. This list may not be exhaustive; indeed, we did not take into account groups of flows

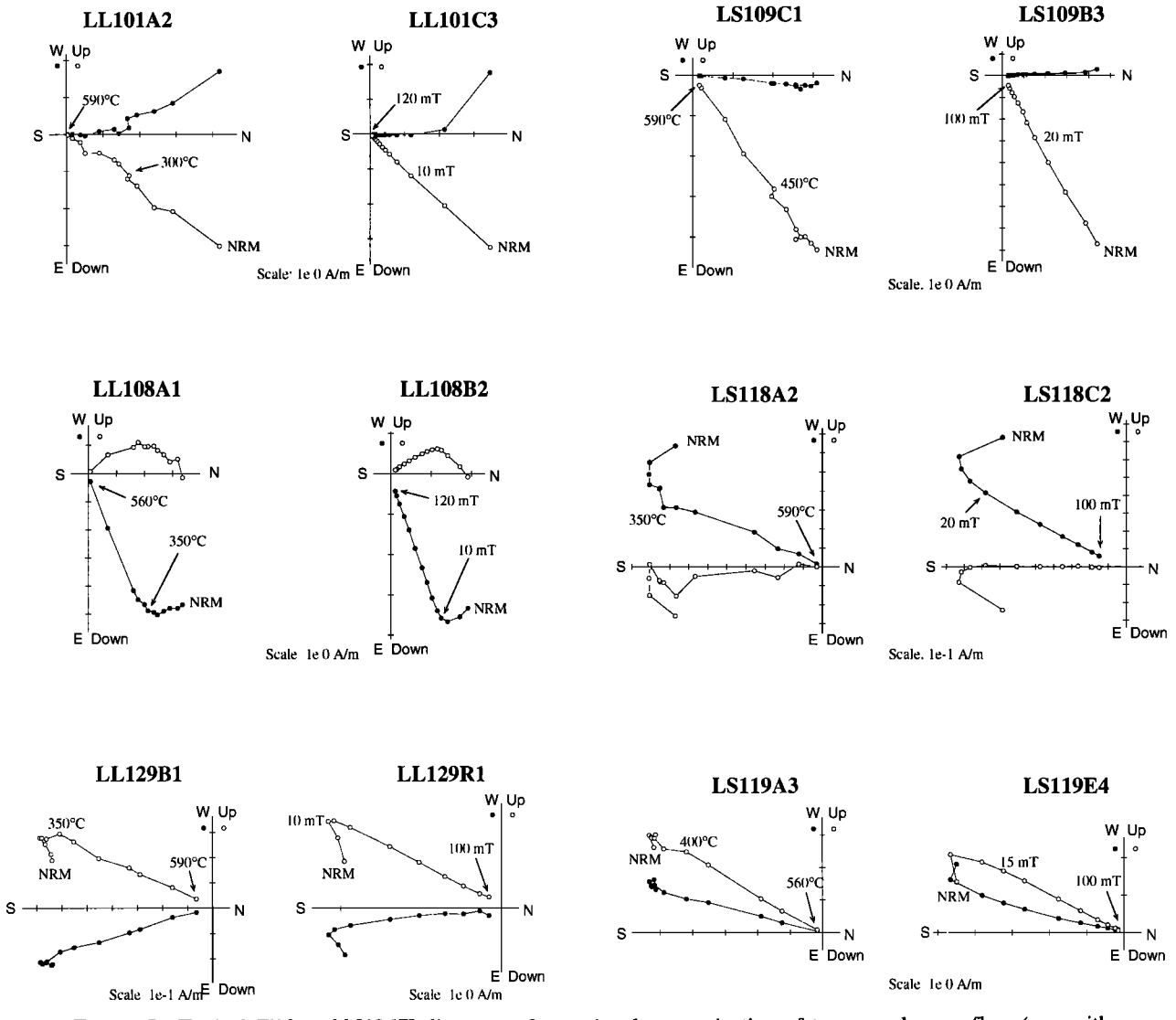


Figure 5. Typical *Zijderveld* [1967] diagrams of stepwise demagnetization of two samples per flow (one with thermal and one with AF treatment) with normal, intermediate, and reverse polarity, from the LL and the LS sections. Solid symbols correspond to the projection onto the horizontal plane, while open symbols are projections onto the vertical plane.

exhibiting the same direction but different mineralogical signatures. In the absence of other criteria, however, it is probably inappropriate to designate other groups. The averaged directions and corresponding VGPs of every group of flows (Tables 1 and 2) is used for analysis and discussion of the results.

Paleointensity

Twenty-three paleointensity determinations from 14 independent groups of flows have been obtained according to the criteria commonly used in palointensity studies [*Nagata et al.*, 1963]. These criteria require NRM-TRM diagrams with linear trends and acceptable pTRM checks. After correcting for the deviations found by the pTRM checks (following the method described above and by *Valet et al.*, [1994b]), we obtained a total of 35 reliable paleointensities from 22 independent groups of flows (Tables 3 and 4). However, acquisition of chemical remanent magnetization (CRM) cannot always be detected by pTRM checks [*Coe et al.*, 1978]. Chemical changes that are produced below a critical temperature may involve conversion of grains with lower (respectively higher) blocking temperatures into grains with higher (respectively lower) blocking temperatures. These altered grains can induce significant changes in the NRM and these changes do not necessarily give negative pTRM checks. Since the field in this study was always directed along the z axis of the samples, any remagnetization subsequent to such chemical changes would tend to align the NRM parallel to the direction of the applied field. Only flows LS117 and LS115 acquired an abnormal z component between 180°C and 320°C.

Half of the NRM-TRM plots display a linear trend with good pTRM checks until relatively high temperatures. A typical sample is shown in Figure 7a together with its corresponding demagnetization diagram. In these cases the pTRM checks are highly reproducible, and the corrected and uncorrected values of paleointensity are basically the same (Table 3 and 4). A large number of NRM-TRM diagrams show non ideal behavior before correction with the linear trend limited to a short temperature interval (Figure 7b). The pTRM corrections reconcile the low and high temperatures by restoring linearity to the NRM-TRM

Table 1. Mean Directions for the LS Section

Site	*N*	Declination, deg	Inclination, deg	α_{95}	*K*	VGP latitude	VGP longitude	*P*
			Time Independent Flows					
LS101	4/4	28.5	45.4	9.0	106	64.8	69.8	N
LS102	3/3	28.5	32.1	13.2	88	61.5	89.6	N
LS103	4/4	4.7	39.2	4.8	366	82.1	128.6	N
LS104	4/4	354.4	32.3	4.2	472	77.6	188.0	N
LS105	4/4	353.5	41.0	3.4	721	82.1	211.5	N
LS106	4/4	12.3	45.8	3.3	754	79.0	77.7	N
LS107	4/4	8.9	49.2	6.6	193	82.1	60.7	N
LS108	4/4	4.4	58.6	5.0	332	78.9	0.2	N
LS109	5/5	5.9	61.3	6.9	125	75.6	0.0	N
LS110	8/9	4.7	51.1	7.2	53	85.0	34.8	N
LS111	5/6	2.8	40.9	4.8	253	84.0	136.6	N
LS112	2/10	143.6	-56.4	N.d.	N.d.	-58.6	96.9	R
LS113	5/5	160.4	-68.5	7.6	103	-62.8	135.3	R
LS114	6/6	144.6	-71.0	11.7	34	-53.3	128.9	R
LS115	7/8	136.1	-74.5	8.4	53	-46.7	132.9	I
LS116	5/5	182.1	-74.6	6.3	150	-57.6	164.2	R
LS117	6/8	172.6	-43.5	10.5	41	-82.6	46.6	R
LS118	6/7	206.3	2.2	3.6	342	-50.9	-62.3	I
LS119	5/5	192.0	-32.0	5.2	214	-74.1	-64.2	R
LS120	5/5	189.9	-33.1	8.1	90	-75.9	-60.0	R
LS121	5/5	166.3	-39.8	6.7	132	-76.2	48.8	R
			Not Time Independent Flows					
LS113/114	11/11	152.0	-70.8	6.2	55	-56.9	132.9	R

N, number of samples used in the analysis/total number of samples treated; α_{95}, 95% confidence cone of the *Fisher's* [1953] statistic, *K*, precision parameter, *P*, polarity of the flow, I, intermediate; N, normal; R, reverse; considered as intermediate when the reversal angle is greater than 30°.

diagrams. This permits unambiguous determinations of paleointensity over a wider range of temperatures and represent a major improvement, especially for nearly transitional samples with a significant normal polarity overprint. An extreme case is shown in Figure 7c. The uncorrected NRM-TRM diagrams are strongly nonlinear, whereas a linear trend is observed at high temperatures after correction. Because of the limited number of pTRM checks, the corrected NRM-TRM diagram is, in this case, defined by only three points, but it corresponds to a wide temperature spectrum, the same, in fact, as the one used for the definition of the ChRM. The corrected NRM-TRM diagrams incorporate more temperature steps than before correction because they restore linearity at high temperatures. The use of high-temperature steps is very important. Indeed, the loss of the ability to acquire a TRM often occurs at low or medium temperatures, while the direction of the chRM is often determined at higher temperatures.

The values of absolute paleointensity determined before and after correction are reported in Tables 3 and 4 using the statistical method developed by *Coe et al.* [1978]. The average paleointensity of each individual flow or group of flows was calculated using a weighting factor *w*, as defined by *Prévot et al.* [1985]. Except for flows LL135 and LS120, we observe good coherency within every individual flow. Note also that the coherency within individual flows has been improved by the correction. The mean paleointensity values range between 3.9 and 46.0 μT. The order of magnitude difference between these two values is in excellent agreement with previous paleointensity determinations involving sequences of transitional and non transitional lava flows [*Prévot et al.*, 1985; *Chauvin et al.*, 1990].

Discussion

Directions that result in VGP latitudes lower than 60° or 45° have usually been considered as transitional. Since the transitional field is probably nondipolar, the use of VGPs is not fully appropriate. *Hoffman* [1984] proposed the use of "rotated directions" (*D'*, *I'*). These are obtained by rotation of the conventional directions (*D*, *I*) about an east-west axis so that the full polarity axial dipole direction has a 90° inclination. Following the approach of *Hoffman* [1984], directions were identified as transitional when the angular distance from the axial dipole was greater than 30°. Only two groups of flows from the LS section and a total of 15 directions in the LL section can be considered as transitional (Tables 1 and 2, respectively).

Transitional Rebound orPostreversal Excursion?

The two VGP paths shown in Figures 8a confirm that the polarity transition is poorly defined in terms of directional changes. The geomagnetic event that follows the transition is better constrained with a cluster of VGPs over northeastern India. In the absence of a detailed chronology, we cannot determine whether this cluster reflects an episode of rapid extrusion or a stationary state of the field. We note that the VGP locations do not coincide with any of the prefered patches identified by

Table 2. Mean Directions for the LL Section.

Site	*N*	Declination, deg	Inclination, deg	α_{95}	*K*	VGP latitude	VGP longitude	*P*
			Time Independent Flows					
LL101	5/7	358.8	40.4	6.1	158	84.2	173.2	N
LL102	7/7	46.8	8.9	11.7	28	39.5	92.0	I
LL103	5/5	61.8	5.5	10.2	57	25.9	84.2	I
LL104	4/4	64.1	-1.3	4.6	396	22.2	86.1	I
LL105	6/6	60.2	-4.7	8.3	67	24.5	89.9	I
LL106	6/7	44.0	8.0	14.3	23	41.5	94.6	I
LL107	4/4	57.4	-2.2	17.7	28	27.6	90.5	I
LL108	4/4	70.9	-13.3	2.7	N.d.	13.2	87.8	I
LL109	4/4	69.8	-10.6	4.2	470	14.9	87.1	I
LL110	4/4	72.9	-6.7	2.0	N.d.	13.2	83.7	I
LL111	4/6	68.9	-17.6	4.4	432	13.7	90.8	I
LL112	7/7	43.7	3.2	16.2	15	40.3	97.4	I
LL113	6/6	52.5	-2.2	10.9	39	31.6	93.6	I
LL114	5/5	42.9	7.6	11.1	48	42.3	95.7	I
LL115	4/4	60.3	-8.3	3.7	604	23.5	91.5	I
LL116	2/6	358.5	-19.9	N.d.	N.d.	50.9	164.6	I
LL117	6/6	17.2	5.1	11.9	33	59.1	127.2	I
LL118	5/5	4.1	32.3	8.9	75	78.1	142.9	N
LL119	4/4	7.5	48.6	6.0	237	83.4	63.9	N
LL120	3/3	2.3	37.2	7.1	304	81.7	147.2	N
LL121	4/4	359.5	37.7	9.0	106	82.3	165.8	N
LL122	4/4	359.9	38.1	5.1	329	82.6	163.0	N
LL123	4/4	356.9	32.8	6.5	203	78.7	177.5	N
LL124	5/5	9.9	28.7	5.6	190	73.7	126.1	N
LL125	5/6	8.7	28.3	3.3	528	74.1	130.1	N
LL126	5/5	9.0	19.0	6.9	122	69.2	136.6	N
LL127	5/6	9.8	35.9	5.1	225	77.4	115.1	N
LL128	8/8	4.5	49.3	5.9	88	85.9	51.9	N
LL129	8/8	157.6	-25.3	11.2	25	-64.1	40.3	R
LL130	5/5	158.3	-14.5	3.4	518	-60.4	30.2	I
LL131	4/4	157.0	-10.1	5.0	336	-57.8	29.2	I
LL132	5/5	144.0	-13.0	9.2	71	-49.4	46.1	I
LL133	5/6	157.1	-19.0	6.5	140	-61.3	35.3	I
LL134	6/6	167.3	-31.7	15.1	21	-73.5	30.1	R
LL135	5/5	185.8	-44.9	5.2	218	-84.4	-84.8	R
LL136	9/9	182.7	-47.1	6.8	59	-87.6	-96.1	R
LL137	4/4	180.4	-40.6	5.7	262	-84.4	-21.5	R
LL138	4/4	205.8	-30.9	10.3	81	-63.4	-86.2	R
LL139	4/4	189.5	-34.8	13.2	50	-77.0	-61.7	R
LL140	3/4	171.4	-32.7	5.9	440	-76.5	19.8	R
LL141	4/4	169.1	-46.8	10.2	81	-80.4	70.3	R
			Not Time Independent Flows					
LL103/105	15/15	61.8	-0.4	4.6	70	24.4	87.0	I
LL109/111	12/14	68.8	-11.6	3.0	207	15.4	88.1	I
LL122/123	8/8	358.4	35.5	3.9	204	80.7	171.7	N
LL124/125	10/11	9.3	28.5	2.8	309	73.9	128.1	N
LL130/131	9/9	157.7	-12.0	2.6	382	-59.0	29.4	I

N, number of samples used in the analysis/total number of samples treated; α_{95}, 95% confidence cone of the *Fisher's* [1953] statistic, *K*, precision parameter; *P*, polarity of the flow, I, intermediate; N, normal; R, reverse; considered as intermediate when the reversal angle is greater than 30°.

Hoffman [1992]. Directional rebounds following reversals have been reported in several sedimentary records of transitions [e.g., *Laj et al.*, 1988], but they have not been observed during the M-B transition. Indeed, the most detailed M-B sedimentary records, from Deep Sea Drilling Project (DSDP) Site 609B [*Clement and Kent*, 1986], Ocean Drilling Project (ODP) Site 664D [*Valet et al.*, 1989], and core V16-58 [*Clement and Kent*, 1991], show longitudinally well-confined VGP paths. Only four volcanic records of the M-B transition have been obtained so far. According to the description of the record from Maui [*Coe et al.*, 1985] (not formally published by the authors and thus to be taken with caution), there are two directions with normal polarity in the

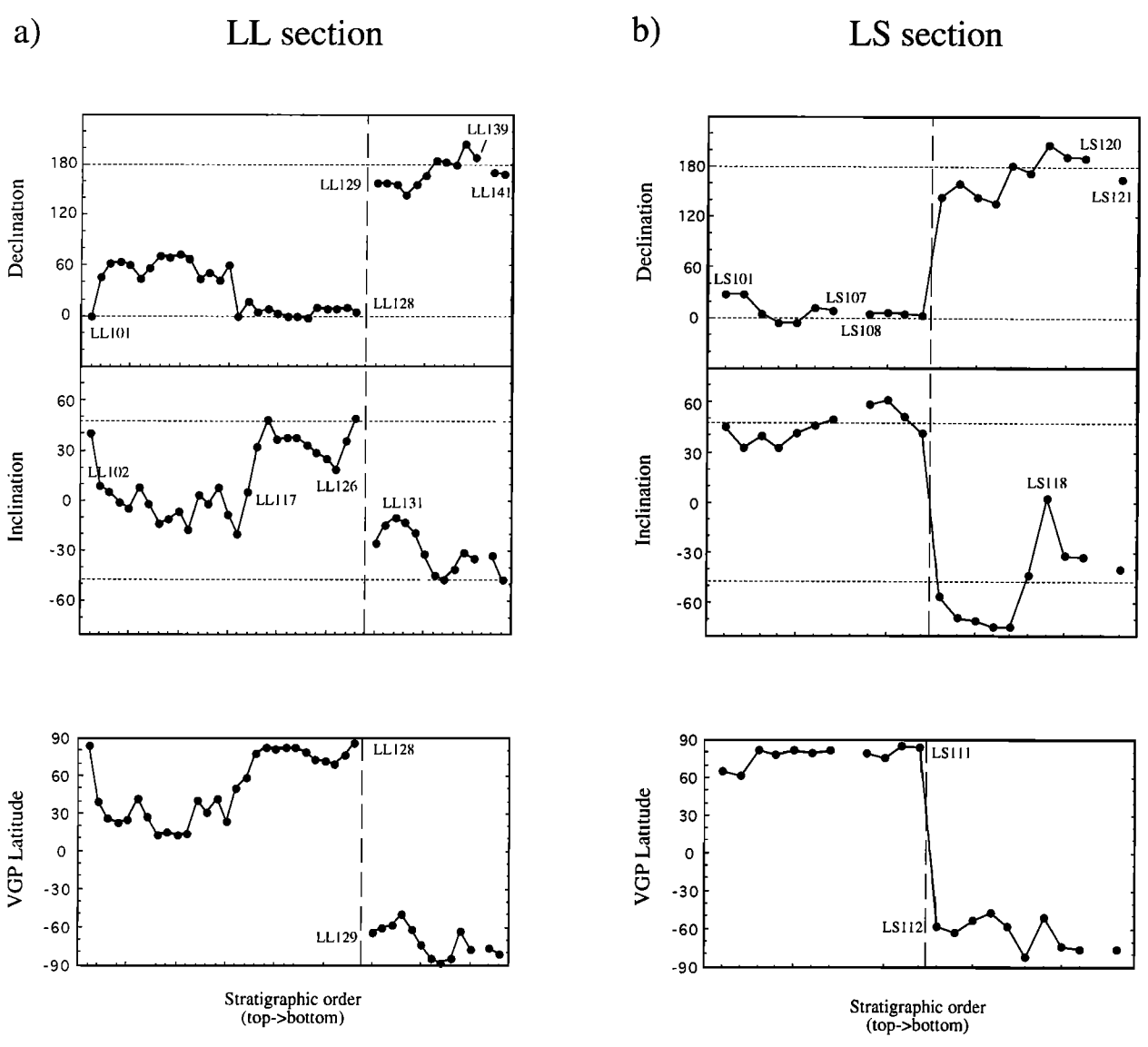


Figure 6. (a) Evolution of the declination, inclination, and VGP latitudes as a function of stratigraphic position in the LL section. The direction of the axial dipole field at La Palma is shown by dashed lines. (b) Same as Figure 6a, but for the LS section.

middle of the transition and thus directly associated with the reversal. This behavior was not seen in a subsequent (but less detailed) record of the same reversal published by *Baksi et al.* [1992]. The M-B volcanic record from Tahiti [*Chauvin et al.*, 1990] is poorly documented and does not show evidence for a postreversal rebound. Thus the hypothesis of a rebound is unlikely, but it cannot be completely ruled out. Further study is needed before a final conclusion can be drawn.

Like the present record, the rebound observed in the 15 Ma R-N reversal from Steens Mountain is separated from the actual transition by an episode of high intensity and full normal directions, which is interpreted as the restoration of a dipolar structure. *Prévot et al.* [1985] favored the hypothesis that both the transition and the rebound are linked to the reversal process. Alternatively, it was proposed [*Valet et al.*, 1985] that the two events are in fact separated by a significant period of time and are therefore disconnected. For the present record of the M-B transition we believe that the restoration of a dipolar structure and the absence of any detailed chronology make both hypotheses equally likely. Evidence for a large deviation from the axial dipole direction in the early part of the Brunhes chron, with an estimated age of 0.62 Ma, was first reported by *Watkins* [1968] from Eltanin core 17-11 and subsequently named the Delta event after a record published by *Creer et al.* [1980]. Four records of the Delta subchron have been reported since then and are summarized by *Champion et al.* [1988]. The best estimate for the age is 0.63-0.64 Ma. An age of 0.69 Ma has recently been reported from a paleointensity low in deep-sea sediments from the equatorial Pacific [*Valet and Meynadier*, 1993]. We note that the intermediate directions recorded in the upper part of the LL section (from LL102 to LL117) (Figure 6a) are also characterized by low field values (from 7.3 μT to 15.0 μT, Table 4). The K/Ar date of 0.78 ± 0.09 Ma for flow LL101 [*Abdel-Monem et al.*, 1972] immediately above the last of these "excursional" flows does not allow determination of whether the rebound can be correlated to the Delta event. New K/Ar age determinations of the excursional flows are in progress, and we hope they will solve this interesting question.

Table 3. Paleointensity Results for the LS section

Sample	D, deg	I, deg	H_{lab}, μT	*Noncorrected*					*Corrected*				
				n	ΔT	w	$H_a \pm \sigma H_a$	$F_a \pm \sigma F_a$	n	ΔT	w	$H_a \pm \sigma H_a$	$F_a \pm \sigma F_a$
LS109Q2	30.2	64.5	21.4						3	309-454	0.52	28.4 ± 5.1	28.4 ± 5.1
LS110B	-	-	38.8	6	297-486	4.48	22.7 ± 0.7		4	269-422	3.15	27.1 ± 0.7	
LS110P2	0.8	32.2	23.2	5	389-514	3.30	19.3 ± 1.1	21.3 ± 1.7	5	186-451	4.68	29.1 ± 0.1	28.3 ± 1.0
LS111Q3	6.2	40.7	23.7	4	345-449	0.64	19.3 ± 2.5	19.3 ± 2.5	3	387-511	0.65	29.5 ± 3.0	29.5 ± 3.0
LS112A3	143.9	-48.6	42.2						3	418-540	0.33	5.5 ± 0.4	5.5 ± 0.4
LS113C2	165.4	-64.8	43.1	5	385-513	1.46	7.1± 0.5		4	354-538	2.21	5.9 ± 0.4	
LS113U2	150.5	-71.4	23.9	11	261-564	10.20	4.7 ± 0.1		5	301-544	13.50	5.9 ± 0.1	
LS114C2	155.8	-53.2	43.9	5	323-445	0.68	4.5 ± 0.5		4	353-536	3.70	3.8 ± 0.2	
LS114S3	115.6	-63.9	23.5	4	340-440	3.61	5.6 ± 0.1	5.1 ± 1.0	4	380-540	3.22	5.0 ± 0.4	5.4 ± 0.9
LS118T2	226.7	-16.5	43.4						4	348-528	0.58	9.1 ± 1.6	
LS118PA2	206.4	-2.4	20.7						3	427-548	1.21	7.2 ± 0.5	7.7 ± 1.0
LS119Q5	185.3	-37.0	22.0						4	393-564	0.59	33.5 ± 4.0	33.5 ± 4.0
LS120B2	189.4	-30.7	40.4						4	346-523	0.38	34.4 ± 3.2	
LS120T2	179.1	-43.7	22.7						3	310-457	10.90	25.1 ± 0.1	25.4 ± 4.7

D, sample declination; I, sample inclination; H_{lab}, laboratory field; n, number of points in the ΔT interval; ΔT, interval of temperature used for the paleointensity determination; w, weighting factor [*Prévot et al.*, 1985]; H_a, paleointensity determination for an individual specimen; σH_a, standard error; F_a, weighted paleointensity average for individual flow or group of flows, σF_a, standard error associated with the weighted average.

Table 4. Paleointensity Results for the LL section

Sample	D, deg	I, deg	H_{lab}, μT	*Noncorrected*					*Corrected*				
				n	ΔT	w	$H_a \pm \sigma H_a$	$F_a \pm \sigma F_a$	n	ΔT	w	$H_a \pm \sigma H_a$	$F_a \pm \sigma F_a$
LL102T3	57.5	-2.3	22.7	4	488-562	0.78	7.7 ± 0.9	7.7 ± 0.9	3	462-562	3.14	7.3 ± 0.4	7.3 ± 0.4
LL103P3	60.0	-2.1	23.2	6	349-514	3.42	7.0 ± 0.2	7.0 ± 0.2	5	265-514	6.91	8.5 ± 0.1	
LL105C2	42.4	-10.5	38.8						3	422-545	0.39	8.2 ± 2.0	8.5 ± 0.2
LL111P3	64.8	-17.2	23.9	7	303-507	5.21	13.4 ± 0.4	13.4 ± 0.4	6	223-507	6.85	15.0 ± 0.4	15.0 ± 0.4
LL113C3	55.6	-10.6	23.0	6	380-520	2.32	7.6 ± 0.5	7.6 ± 0.5	4	380-540	4.00	9.2 ± 0.4	9.2 ± 0.4
LL121P2	354.2	28.8	22.0						6	190-460	1.62	46.0 ± 3.6	46.0 ± 3.6
LL128F2	6.5	44.0	40.6						4	357-542	0.65	41.2 ± 5.4	41.2 ± 5.4
LL129Q3	118.2	-44.4	23.7	5	388-511	1.16	2.4 ± 0.3		4	385-552	1.20	4.7 ± 0.5	
LL129S2	153.9	-23.4	23.9	8	181-442	1.59	9.6 ± 0.7	6.6 ± 3.6	7	181-504	7.69	8.8 ± 0.2	8.2 ± 2.0
LL130C1	154.5	-8.5	44.0	5	381-504	1.90	13.2 ± 0.6		4	351-530	15.60	11.1 ± 0.1	
LL130Q3	158.6	-15.0	20.7	5	335-462	1.77	12.2 ± 0.7		8	178-528	7.46	8.6 ± 0.2	
LL131Q2	153.7	-11.5	18.9	5	370-486	1.30	9.7 ± 0.9	11.9 ± 1.5	4	370-526	4.89	6.9 ± 0.2	9.7 ± 1.7
LL132B2	132.1	-12.2	43.4	4	318-408	0.24	5.9 ± 0.7		6	181-408	0.92	3.9 ± 0.2	
LL132P2	141.8	-14.2	18.9	7	370-524	2.57	3.6 ± 0.2	3.8 ± 1.1	7	177-486	1.86	3.9 ± 0.2	3.9 ± 0.0
LL133R3	154.9	-14.1	23.0	5	410-520	1.19	4.7 ± 0.5	4.7 ± 0.5	4	380-540	0.89	5.9 ± 0.9	5.9 ± 0.9
LL134A3	147.5	-37.9	43.1	7	225-416	0.98	2.1 ± 0.2		7	225-520	2.00	8.2 ± 0.7	
LL134P3	172.0	-34.6	23.9	8	303-528	8.13	6.2 ± 0.2	5.8 ± 2.0	4	303-507	3.85	6.4 ± 0.4	7.0 ± 0.9
LL135F3	174.2	-42.2	43.9	6	293-445	1.18	40.8 ± 3.7	40.8 ± 3.7	8	180-535	2.11	47.0 ± 3.9	
LL135E3	184.1	-45.0	22.0						3	434-532	0.37	21.8 ± 3.4	43.3±12.6
LL138D2	218.7	-38.5	23.5						3	301-441	2.91	22.5 ± 1.1	22.5 ± 1.1
LL139S3	184.9	+33.5	23.5	4	300-410	1.26	30.8 ± 2.0	30.8 ± 2.0	7	181-503	3.44	24.8 ± 1.4	24.8 ± 1.4

D, sample declination; I, sample inclination; H_{lab}, laboratory field, n, number of points in the ΔT interval; ΔT, interval of temperature used for the paleointensity determination; w, weighting factor [*Prévot et al*., 1985], H_a, paleointensity determination for an individual specimen, σH_a, standard error, F_a, weighted paleointensity average for individual flow or group of flows; σF_a, standard error associated with the weighted average.

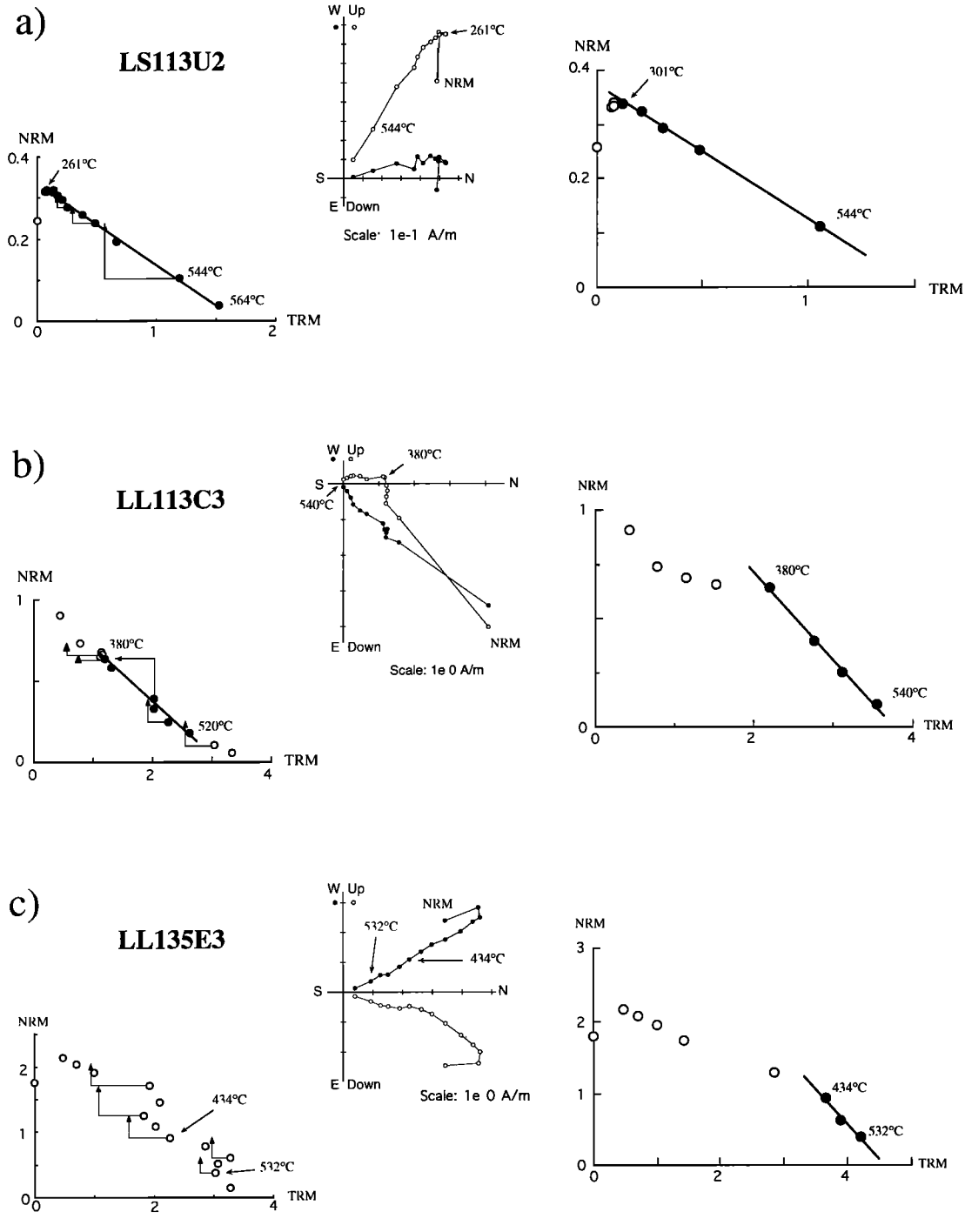


Figure 7. NRM-TRM diagrams (left) before and (right) after correction from magnetomineralogical changes using the pTRM checks, for samples (a) LS113U2, (b) LL113C3, and (c) LL135E3. The intensity values are in A m^2 kg^{-1}. The distinct components of magnetization can be observed in the Zijderveld diagrams (sample coordinates). The slope of the NRM-TRM curves are obtained from regression lines through the data points shown with solid symbols.

Long Lived Transitional States or Episodes of High Extrusion Rate?

The VGP positions and the available paleointensity values from the four distinct volcanic records of the M-B transition (including the present study with corrected paleointensities) are plotted in Figure 9. The most detailed record from the two studies in Maui [*Coe et al.*, 1985] is not published and therefore could not be included. The other study contributes four VGPs [*Baksi et al.*, 1992] plotted in Figure 9. Two of these are located over southeast America and fall roughly within a cluster identified by *Hoffman* [1992], while the two others follow a mid-Pacific meridian. The cluster of VGPs close to and north of Australia was already noticed by *Brown et al.* [1994] and was interpreted by them as evidence that the field retains a strong dipole component during the B-M transition. In fact ,this cluster encompasses VGPs from two studies, the Chilean Andes record [*Brown et al.*, 1994], and

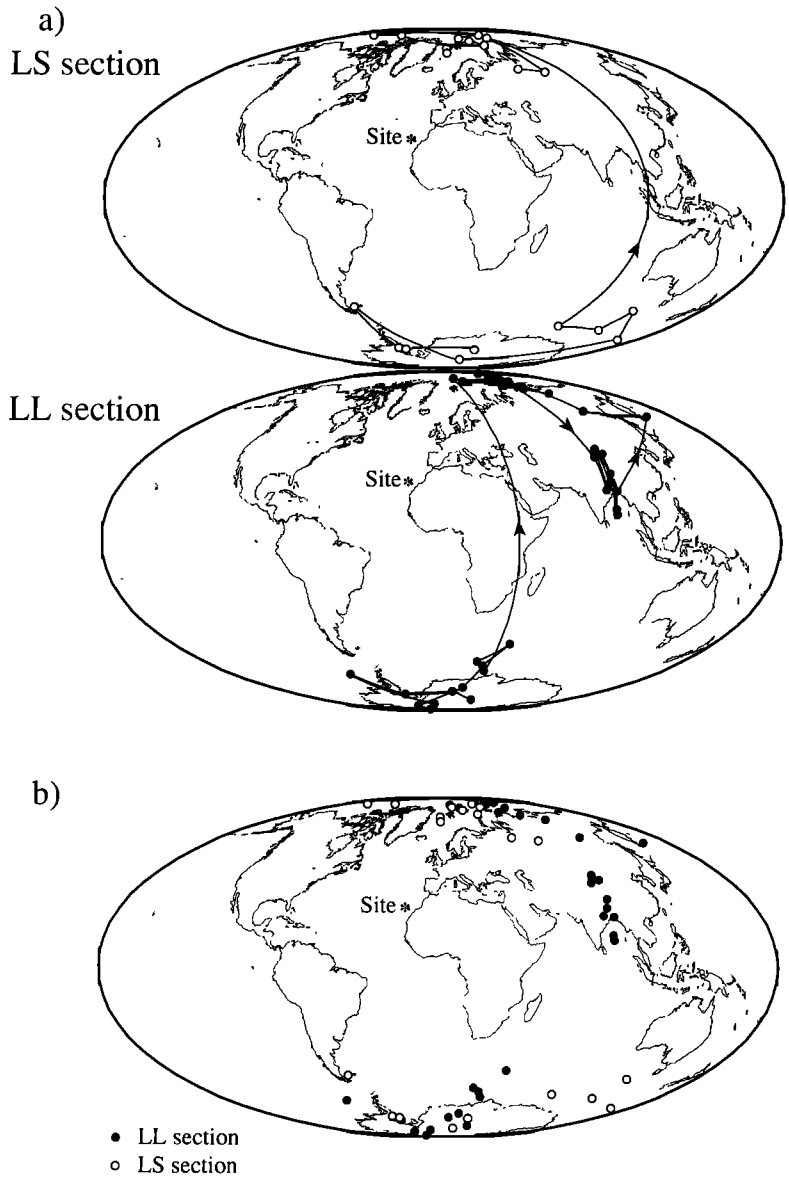


Figure 8. (a) Virtual geomagnetic pole (VGP) paths for the LS and LL sections. (b) Individual VGPs from the LS (open symbols) and LL (solid symbols) sections.

the Tahiti record [*Chauvin et al.*, 1990]. The individual directions are very tightly grouped, but the two groups do not overlap. We emphasize also that the two unique paleointensity determinations from Tahiti give very low values (3.4 μT and 3.6 μT) (Figure 9) which do not seem to be consistent with the persistence of a strong dipole field component during the reversal. We are thus more inclined to favor the hypothesis of a fortuitous similarity between the two directional groups caused by short episodes with high extrusion rates.

Asymetrical Saw-Tooth Pattern?

Figure 10 shows the absolute paleointensity determinations for the two sections in this study and their relationship to the VGP latitudes. For comparison we plotted the results obtained before and after corrections, which basically depict the same features. The lowest flows of the LS section exhibit high paleointensity and reverse polarity, but the "pretransitional" flows are clearly characterized by low intensities and reverse polarity. Three high field values are observed after the polarity change. In both sections the pretransitional and posttransitional values show very similar characteristics. Indeed, in both sections, low intensities (3.9 μT to 9.7 μT) are recorded immediately before the transition, while normal polarity flows exhibit high field strengths. In the absence of a detailed chronology, it is not possible to conclude whether such intensity variations across the reversal boundary represent the asymmetrical pattern that is observed in sediments of the Pacific, the Atlantic, and the Indian Oceans [*Meynadier et al.*, 1994; *Valet et al.*, 1994a]. However, the present data set does not contradict this observation. Indeed, both sections show evidence for strong intensity recovery immediately following the reversal, whereas the pretransitional period is characterized by low values. A similar asymmetry can be observed in the detailed record from Steens Mountain [*Prévot et al.*, 1985]. Strong and probably rapid field recovery following a transition was also reported on Kauai [*Bogue and Paul*, 1993] and in the record of the upper Jaramillo subchron on Tahiti [*Chauvin et al.*, 1990]. As

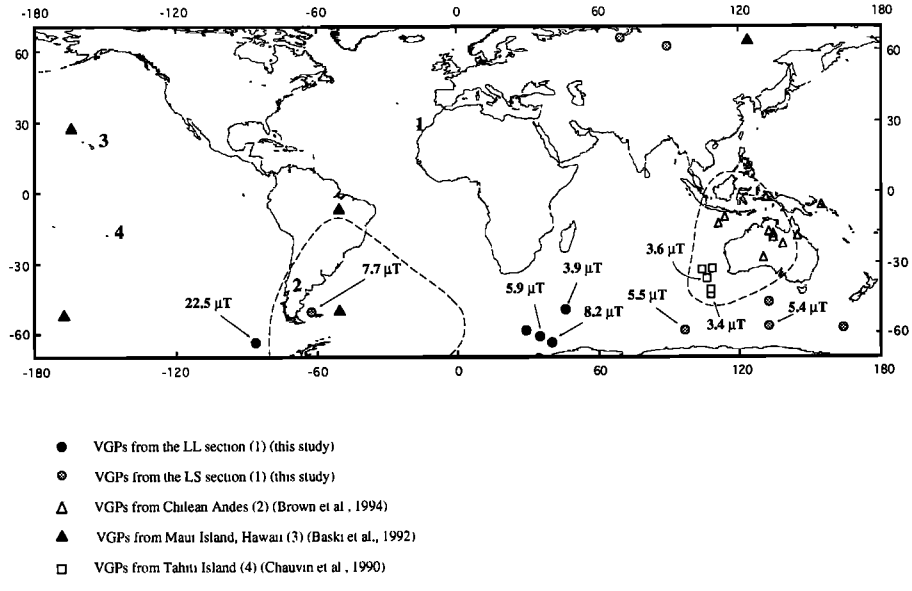


Figure 9. Compilation of the individual VGPs from the published volcanic records of the Brunhes-Matuyama transition and those of the present study. Also shown are the locations of the four sampling sites. The absolute paleointensities, when available, have been plotted with their corresponding VGP. The locations of the VGP clusters as defined by *Hoffman* [1992] are shown by dashed lines.

already emphasized by *Bogue and Paul* [1993], these results indicate that a strong posttransitional dipolar field could well be a systematic feature of the reversing field.

Transitional Field Intensities

A model recently proposed by *Gubbins* [1994] and *Gubbins and Coe* [1993] incorporates growth and drift of positive and negative flux at the core surface. They argue that under these circumstances the VGP paths can be confined relatively close to a single longitude despite the presence of large nondipole fields. The transitional stage is in fact mostly controlled by an equatorial dipole, which surprisingly reduces to zero during the prereversal and postreversal stage. The model implies the existence of high transitional field intensities which are not observed in the four available records of the M-B transition.

In Figure 11, we plot the absolute paleointensities (obtained after correction) of the M-B records from Tahiti and La Palma as a function of the angle to the normal axial field direction. As was noted recently [*Lin et al.*, 1994], large changes in paleointensity occur while the pole is still at high latitudes. Since we are dealing with the total field intensity, this behavior could reflect a decrease of both the dipole and nondipole components. However, owing to the short-term constants of the nondipole field [*Hulot and Le Mouël*, 1994], it is difficult to imagine that the ratio between the axial dipole and the other terms would remain unaffected during the entire reversal process. Indeed, if that were the case, the geomagnetic field would remain dipolar, and there would not be any transitional directions. Instead, paleomagnetic observations from sedimentary sequences indicate a long-term decline over a period of more than several thousand years. This is difficult to reconcile with the time constants of the nondipole field. We prefer to interpret this behavior as the consequence of a systematic decrease of the dominant dipole component, which at some point favors the emergence of the nondipole components. The transitional and/or excursional paleointensities (Table 3 and 4) range between 3.9 μT and 15.0 μT with a mean value of 8 ± 3 μT, in agreement with those reported in other studies (e.g., 5.5 μT in sequences from Iceland [*Lawley*, 1970]; 5 μT from young reversals at Tahiti [*Chauvin et al.*, 1990; 11 μT at the Steens Mountain [*Prévot et al.*, 1985]. At the present time, the nondipole field (1985 International Geomagnetic Reference Field (IGRF)) ranges from 1 μT to 16 μT with an average value of 8 μT (± 5 μT) at the latitude of La Palma compared to a value of 9 μT from the statistical model of *Constable and Parker* [1988]. These results are consistent with the hypothesis that the transitional stage is controlled by nondipole components. There is no evidence for the high paleointensities associated with intermediate directions that are needed to support the hypothesis of intermediate dipolar reversal states.

Observations Versus Predictions of a Simple Model

From these observations, we turned back to the simplest model for a geomagnetic field reversal, namely, a decrease of the dipole to zero and its subsequent recovery with the opposite polarity [*Courtillot et al.*, 1992]. We examined the basic observations that would emerge from such a model in terms of the distribution of VGPs and the intensity variations. For the sake of simplificity, we used a standing nondipole field as defined by the Gauss coefficients of the 1985 IGRF. Time-varying components were not included since we have no idea of their time constants with respect to the duration of the reversal process. Previous results from a statistical model [*Constable*, 1990] show that the presence of such components would simply introduce scatter between the successive directions. A total of 101 successive samplings of the geomagnetic field were made between the initial and final intensity values. This procedure was repeated at the four observation sites of the available volcanic records of the M-B transition (Figure 9) (although any other site could have been used as well). The pattern of successive VGP positions deduced from the calculation (Figure 12) are very different at each site. In two

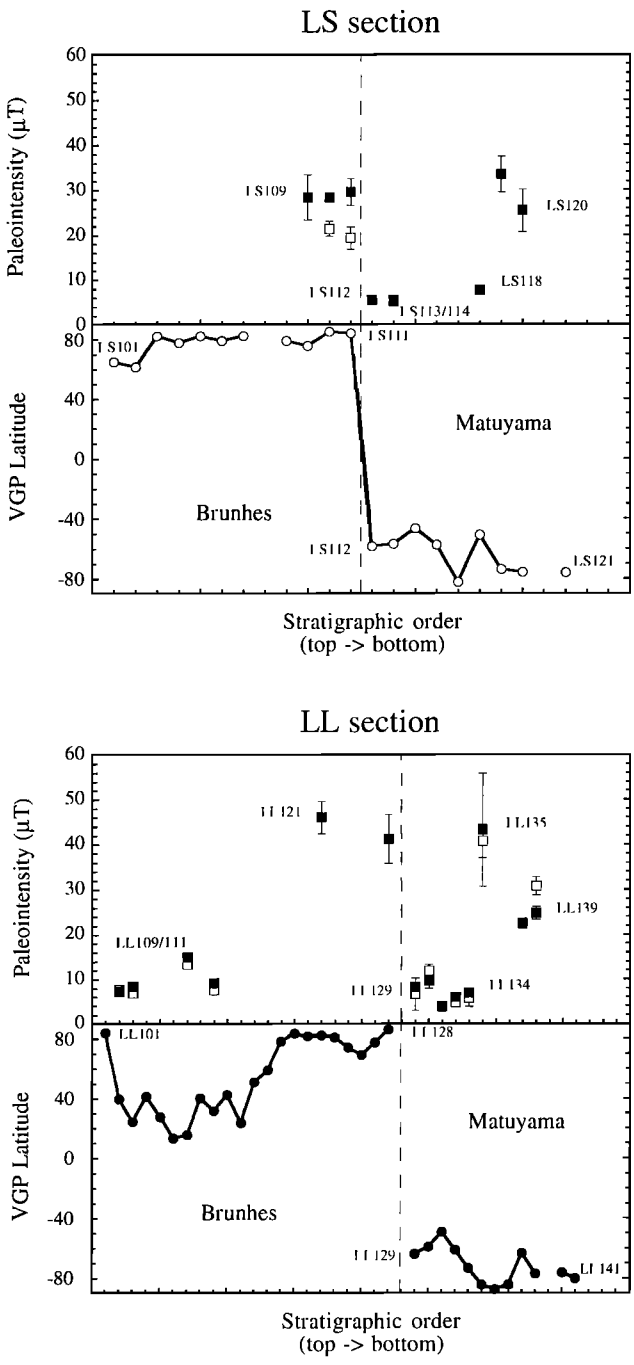


Figure 10. Evolution of the absolute paleointensity and VGP latitude of flows from the (top) LS and (bottom) LL sections. The corrected mean paleointensity value of each flow or group of flow is shown with solid symbols, while the noncorrected value, when available, is shown with open symbols. Error bars are at one-sigma uncertainty. The dashed line shows the position of the reverse-to-normal polarity transition within each section.

cases the zero dipole field intensity corresponds to VGP positions at high northern and southern latitudes (sites 2 and 4), while in the other two cases the coevals VGPs are found close to the equator (sites 1 and 3). The number of intermediate VGP positions also differs considerably (from 2 to more than 30) so that the apparent duration of the transition is strikingly different at each site. In Figure 12 b we plot the intensity of the magnetic field as a function of the VGP latitudes at the four sites. (Identical results would have been obtained using the reversal angle instead of the VGP latitude.) The transitional intensities are between one fifth and one tenth of the prereversal or postreversal values at the sites of La Palma and Chile, which is compatible with the paleomagnetic data. At the other two sites (Maui and Tahiti), the decrease appears to be more pronounced with values down to 1μT. This is lower than most transitional paleointensities but not in contradiction with the lowest values (3.4 μT at Tahiti). In any case, it should be noted that crustal or local fields probably play a significant role when the geomagnetic field has such a low strength. A characteristic common to the four simulations is the relative absence of VGPs at very low latitudes (between 30°S and 30°N). In fact, in this case, the concept of transitional directions may be meaningless since any distribution of the VGPs is possible, and there is the appearance of rapid geomagnetic field changes. It is thus not surprising that we do not observe coherent and reproducible features among the four individual records of the M-B transition, although they all show evidence for low transitional paleointensities. The preponderance of VGPs in the southern hemisphere during the M-B reversal (Figure 9) could simply be simulated by introducing an asymmetrical evolution of the field intensity, similar to the recent observations in sediments [*Valet and Meynadier*, 1993; *Meynadier et al.*, 1994; *Valet et al.*, 1994a]. These observations emphasize that detailed records of paleointensity will be most crucial in further documenting the evolution of the field during reversals.

Conclusion

The major characteristics that emerge from the present study and from the comparison with other volcanic records of the M-B transition can be summarized as follows:

1. There are fewer low as opposed to high latitude intermediate VGP positions.

2. There is a preponderance of VGPs in the southern hemisphere.

3. The intermediate VGPs are characterized by paleointensities between 5 and 10 times lower than the prereversal or postreversal values.

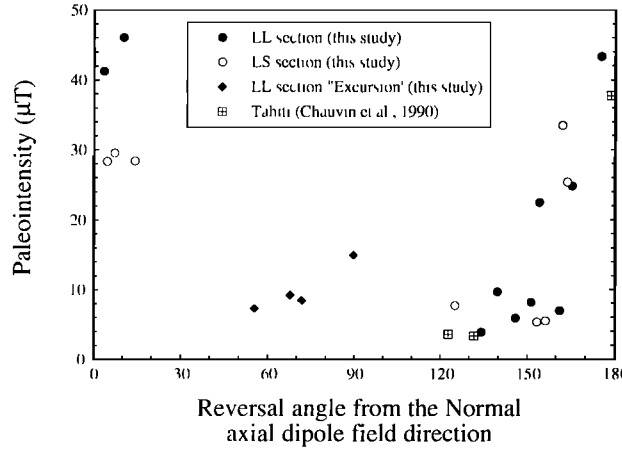


Figure 11. Absolute paleointensities available for the Brunhes-Matuyama transition versus reversal angle (angle between the mean flow direction and the axial dipole direction calculated at the sampling site). Data from the rebound (or the excursion) are shown separately.

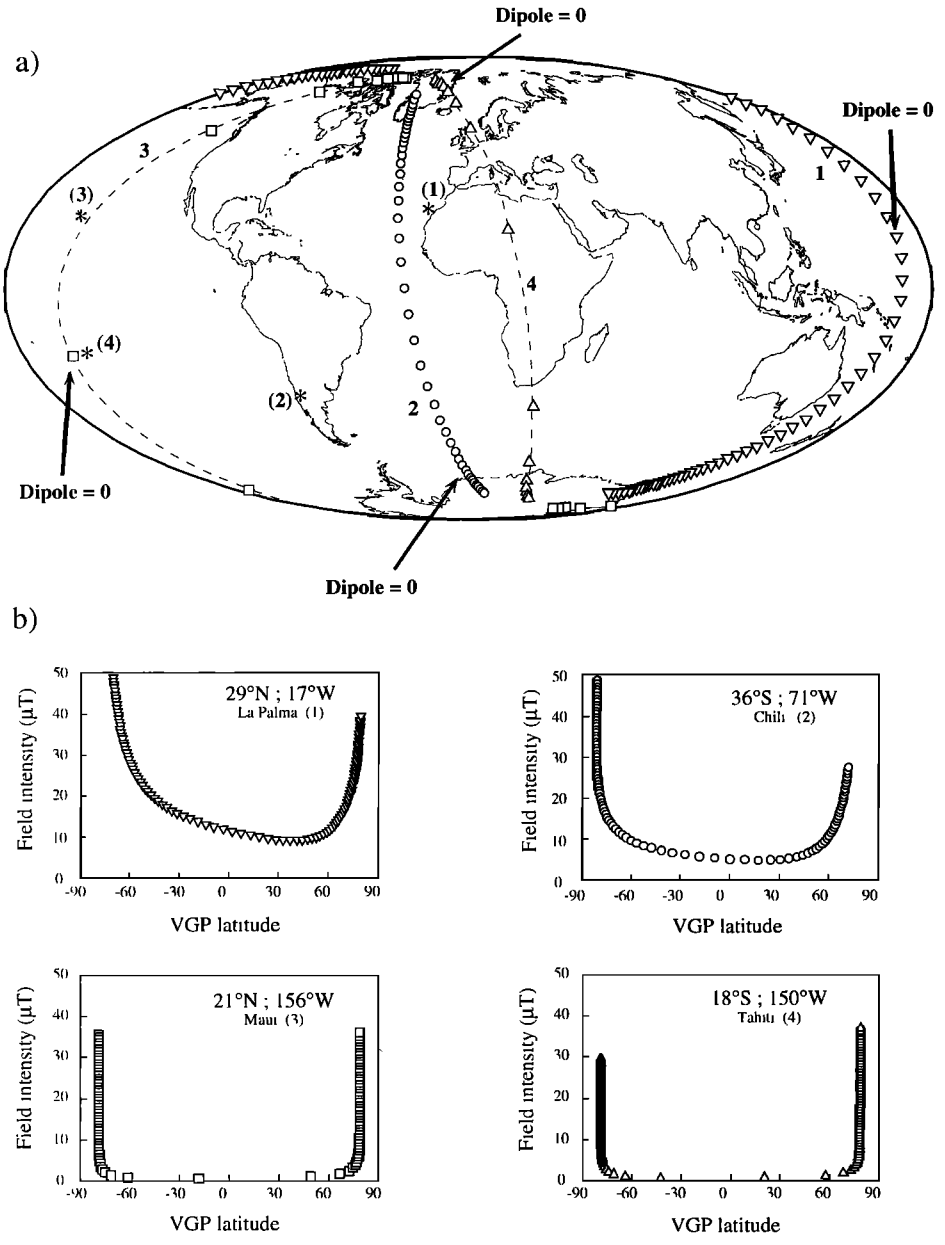


Figure 12. (a) Simulation of VGP paths at four observation sites corresponding to the volcanic records of the Matuyama-Brunhes transition. (b) Field intensity versus VGP latitude at the same sites predicted by the same model. Note the very large differences between sites.

4. The intensity patterns preceding and following the reversal do not seem to be symmetric.

These characteristics can be compared with the predictions of a simple model that involves the linear decay of a dipole superimposed on the nondipole field of the 1985 IGRF and the subsequent recovery of that dipole in the opposite polarity. We realize that the hypothesis of a standing nondipole field is not realistic as there is evidence for drifting and rapidly changing nondipole terms [*Hulot and Le Mouël,* 1994]. However, some insight can be obtained about the existing paleomagnetic records. In particular, every site has different VGP paths, the distributions of the VGP positions differ markedly, and the transitional field intensity can be as low as 4 μT even while the VGP is still at high latitude.

We thus believe that the present transitional data set can be interpreted in terms of the emergence of time varying nondipole components which are favored by the decrease and subsequent recovery of the dipole to the opposite polarity.

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